The documentation and process conversion measures necessary to comply with this document shall be completed by 27 December 2022.

INCH-POUND

MIL-PRF-19500/545L
w/AMENDMENT 1
27 September 2022
SUPERSEDING
MIL-PRF-19500/545L
7 October 2016

PERFORMANCE SPECIFICATION SHEET

# TRANSISTOR, PNP, SILICON, POWER, ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT PACKAGES) AND UN-ENCAPSULATED (DIE), TYPES 2N5151, 2N5153, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, power transistors for use in high-speed power-switching applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type and Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type as specified in MIL-PRF-19500. Provisions for radiation hardness assurance (RHA) to eight radiation levels ("M", "D", "P", "L", "R", "F", "G", and "H") are provided for JANTXV and JANS product assurance levels.

1.2 Package outlines. The device package outlines are as follows: similar to TO-205 in accordance with figure 1 and U3 in accordance with figure 2 for all encapsulated device types. See figures 3, 4, and 5 for unencapsulated devices.

1.3 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Types | $P_T$ $T_A$ = +25°C (1) | $P_T$ $T_C$ = +25°C (1) | $R_{\theta JA}$ (2) | $R_{\theta JC}$ (2) | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $I_C$ (3) | Reverse pulse energy (4) | $T_{STG}$ and $T_J$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | °C/W | V dc | V dc | V dc | A dc | A dc | mj | °C |
| 2N5151, L | 1 | 10 | 175 | 10 | -100 | -80 | -5.5 | -2 | -10 | 15 | -65 to + 200 |
| 2N5153, L | 1 | 10 | 175 | 10 | -100 | -80 | -5.5 | -2 | -10 | 15 | |
| 2N5151U3 | 1.16 | 100 | 150 (5) | 1.75 | -100 | -80 | -5.5 | -2 | -10 | 15 | -65 to + 200 |
| 2N5153U3 | 1.16 | 100 | 150 (5) | 1.75 | -100 | -80 | -5.5 | -2 | -10 | 15 | |

(1) See figures 6, 7, 8, and 9 for temperature-power derating curves.
(2) See figures 10, 11, and 12 for transient thermal impedance graph.
(3) This value applies for Pw ≤ 8.3 ms, duty cycle ≤ 1 percent.
(4) This rating is based on the capability of the transistors to operate safely in the unclamped inductive load energy test circuit, see subgroup 5 of the group A inspection table and figure 13.
(5) Mounted on an FR4 printed circuit board.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Limits | $h_{FE2}$ (1) $V_{CE}$ = -5 V $I_C$ = -2.5 A dc | | $\|h_{fe}\|$ $V_{CE}$ = -5 V $I_C$ = -500 mA dc f = 10 MHz | | $V_{BE(sat)2}$ (1) $I_C$ = -5 A dc $I_B$ = -500 mA dc | $V_{CE(sat)2}$ (1) $I_C$ = -5 A dc $I_B$ = -500 mA dc | $C_{obo}$ $V_{CB}$ = -10 V dc $I_E$ = 0 f = 1 MHz |
|---|---|---|---|---|---|---|---|
| | 2N5151 (2) | 2N5153 (2) | 2N5151 (2) | 2N5153 (2) | | | |
| | | | | | V dc | V dc | pF |
| Min | 30 | 70 | 6 | 7 | | | |
| Max | 90 | 200 | | | 2.2 | 1.5 | 250 |
| (TO-205) Max (U3) | 90 | 200 | | | 2.2 | 1.5 | 250 |

(1) Pulsed, see 4.5.1.
(2) The limits specified apply to all package outlines unless otherwise stated.

* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H").

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "5151" and "5153".

1.5.3.3 Suffix letters. The suffix letter "L" is used on devices that are packaged in the TO-205 package of figure 1 that have a long lead length: 1.500 inches (38.10 mm) minimum and 1.750 inches (44.45 mm) maximum. Devices with no suffix have standard length leads of 0.5 inch (12.7 mm) minimum to .75 inch (19.1 mm) maximum in figure 1. The suffix letters "U3" are used on devices that are packaged in the surface mount package of figure 2.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "B", "C" and "D".



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* 1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

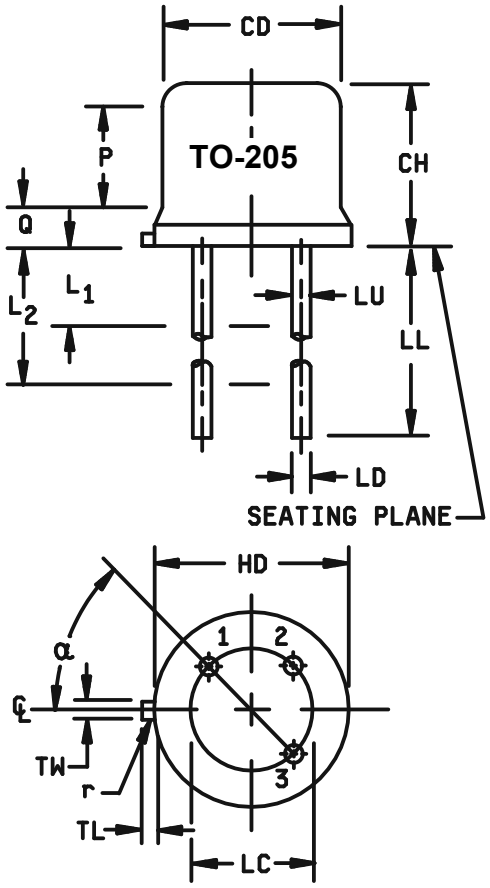
For device type 2N5151, 2N5151L, AND 2N5151U3:.....300 krads(Si) 1/
For device type 2N5153, 2N5153L, AND 2N5153U3:.....300 krads(Si) 1/

* 1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 300 krads(Si).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | IN | IN | mm | mm | |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | 6 |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LC | .200 TP | | 5.08 TP | | 7 |
| LD | .016 | .021 | 0.41 | 0.53 | 8, 9 |
| * LL | | | | | 8, 9, 13, 14 |
| LU | .016 | .019 | 0.41 | 0.48 | 8, 9 |
| $L_1$ | | .050 | | 1.27 | 8, 9 |
| $L_2$ | .250 | | 6.35 | | 8, 9 |
| Q | | .050 | | 1.27 | 6 |
| TL | .029 | .045 | 0.74 | 1.14 | 4, 5 |
| TW | .028 | .034 | 0.71 | 0.86 | 3 |
| r | | .010 | | 0.25 | 11 |
| $\alpha$ | 45° TP | | 45° TP | | 7 |
| P | .100 | | 2.54 | | |

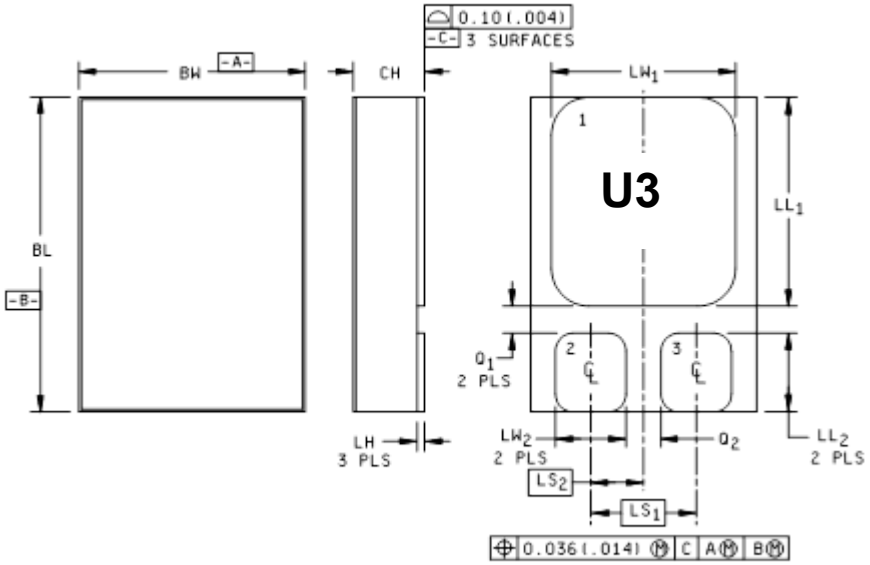



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
8. LU applied between $L_1$ and $L_2$. LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
9. All three leads.
10. The collector shall be electrically and mechanically connected to the case.
11. r (radius) applies to both inside corners of tab.
12. * In accordance with ASME Y14.5, diameters are equivalent to $\phi$x symbology.
13. For transistor types 2N5151 and 2N5153, LL is .5 inch (13 mm) minimum, and .75 inch (19 mm) maximum.
14. For transistor types 2N5151L and 2N5153L, LL is 1.5 inch (38 mm) minimum and 1.75 inch (44.4 mm) maximum.
15. Lead designation, depending on device type, shall be as follows: lead numbering; lead 1 = emitter, lead 2 = base, and lead 3 = collector.

FIGURE 1. Physical dimensions (TO-205).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | IN | IN | mm | mm |
| | Min | Max | Min | Max |
| BL | .395 | .405 | 10.04 | 10.28 |
| BW | .291 | .301 | 7.40 | 7.64 |
| CH | .1085 | .1205 | 2.76 | 3.06 |
| LH | .010 | .020 | 0.25 | 0.51 |
| LL1 | .220 | .230 | 5.59 | 5.84 |
| LL2 | .115 | .125 | 2.93 | 3.17 |
| LS1 | .150 BSC | | 3.81 BSC | |
| LS2 | .075 BSC | | 1.91 BSC | |
| LW1 | .281 | .291 | 7.14 | 7.39 |
| LW2 | .090 | .100 | 2.29 | 2.54 |
| Q1 | .030 | | 0.762 | |
| Q2 | .030 | | 0.762 | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Terminal 1 - collector, terminal 2 - base, terminal 3 - emitter.

SCHEMATIC

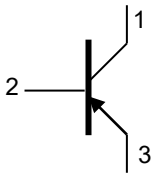
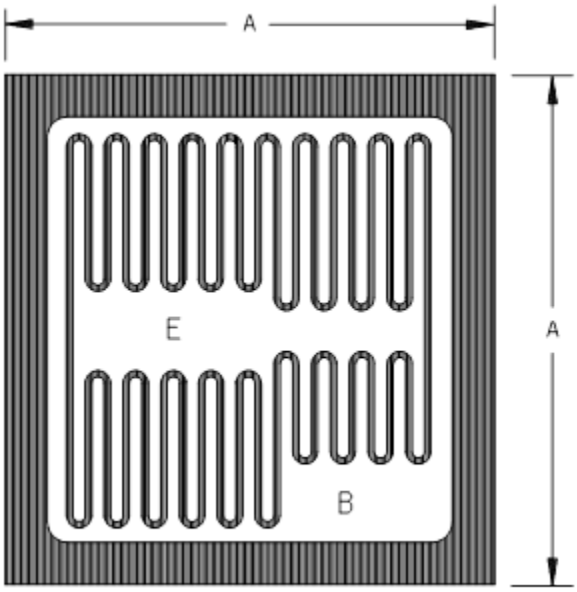



FIGURE 2. Physical dimensions and configuration for surface mount (U3).



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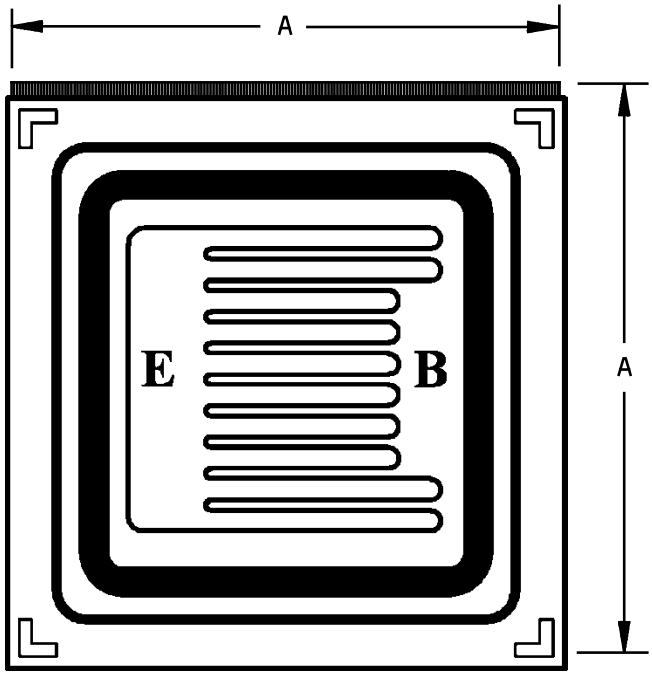

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | in | in | mm | mm |
| | Min | Max | Min | Max |
| A | .100 | .105 | 2.54 | 2.67 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are:
   Thickness: .0078 inch (0.198 mm) nominal, tolerance is ±.005 inch (0.13 mm).
   Top metal: Aluminum, 25,000 Å minimum, 33,000 Å nominal.
   Back metal: Gold 1,500 Å minimum, 2,500 Å nominal.
   Back side: Collector.
   Bonding pad: .012 inch (0.305 mm) min. x .030 inch (0.761 mm) minimum.

FIGURE 3. JANHC and JANKC (B-version) die dimensions.



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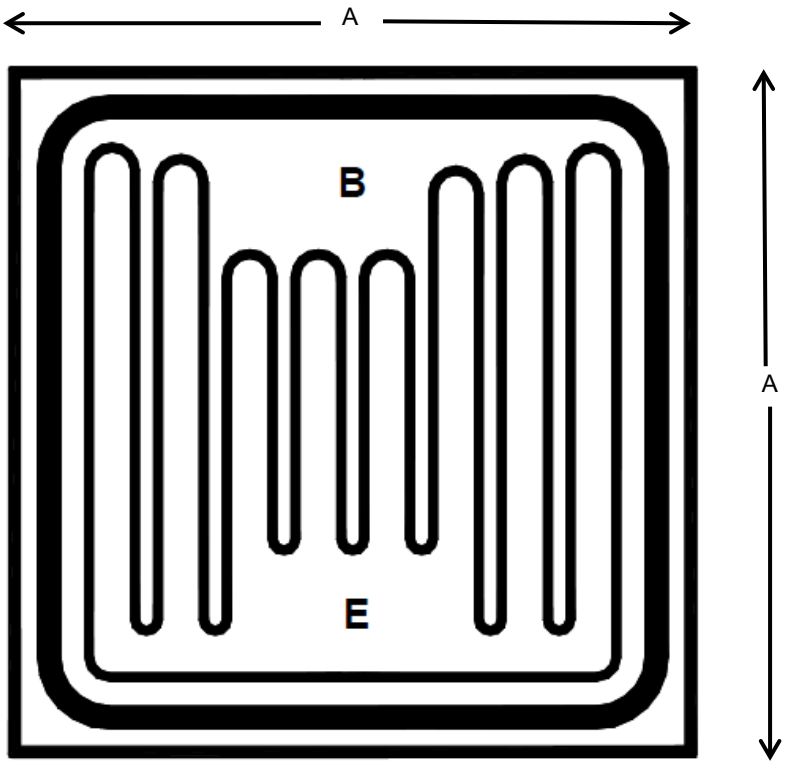

| Ltr. | Dimensions | | | |
|---|---|---|---|---|
| | IN | IN | mm | mm |
| | Min | Max | Min | Max |
| A | .126 | .130 | 3.20 | 3.30 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .010 inch (0.25 mm) ±.0015 inch (0.038 mm) nominal.
   Top metal: Aluminum 30,000 Å minimum, 33,000 Å nominal.
   Back metal: A. Al/Ti/Ni/Ag15kÅ/2kÅ/7kÅ/7kÅ min. 18kÅ/3kÅ/10kÅ/10kÅ nom.
   B. Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: .012 inch (0.305 mm) min. x .030 inch (0.761 mm) minimum.

FIGURE 4. JANHC and JANKC (C-version) die dimensions.



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**Backside: Collector**

| Dimensions | | | | |
|---|---|---|---|---|
| LTR | IN | IN | mm | mm |
| | Min | Max | Min | Max |
| A | .118 | .122 | 3.0 | 3.1 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. The physical characteristics of the die are:
   - Thickness: .0135 inch (0.34 mm) nominal, tolerance is ±.0015 inch (0.04 mm).
   - Top metal: Aluminum, 54,000 Å minimum, 60,000 Å nominal.
   - Back metal: Gold 6,400 Å minimum, 8,000 Å nominal.
   - Back side: Collector.
   - Bonding pad: B = .038 x .022 inch (0.97 x 0.56 mm)
     E = .042 x .020 inch (1.07 mm x 0.51 mm)

FIGURE 5. JANHC and JANKC (D-version) die dimensions.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil).

* 2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* THE AMERICAN SCIETY OF MECHANICAL ENGINEERS (ASME)

* ASME Y14.5 - Dimensioning and Tolerancing

(Copies of these documents are available online at https://www.asme.org )

* 2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to TO-205), figure 2 (U3), and figures 3, 4, and 5 (die dimensions) for JANHC and JANKC.



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3.4.1 Current density. Current density of internal conductors shall be as specified in MIL-PRF-19500.

3.4.2 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I, II, III, and IV).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table IV tests, the tests specified in table IV herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurements for JANS levels | Measurements for JANTX and JANTXV levels |
|---|---|---|
| 3b | Not applicable | Not applicable |
| (1) 3c | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3) | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3.) |
| 9 | $I_{CES1}$ and $h_{FE2}$ | Not applicable |
| 10 | 48 hours minimum. | 48 hours minimum. |
| 11 | $I_{CES1}$ and $h_{FE2}$; $\Delta I_{CES1}$ = 100 percent of initial value or -100 nA dc, whichever is greater. $\Delta h_{FE2}$ = ±20 percent | $I_{CES1}$ and $h_{FE2}$ |
| 12 | See 4.3.2 | See 4.3.2 |
| 13 | Subgroups 2 and 3 of table I herein; $\Delta I_{CES1}$ = 100 percent of initial value or -100 nA dc, whichever is greater. $\Delta h_{FE2}$ = ±20 percent | Subgroup 2 of table I herein; $\Delta I_{CES1}$ = 100 percent of initial value or -100 nA dc, whichever is greater. $\Delta h_{FE2}$ = ±20 percent |

(1) Shall be performed anytime after temperature cycling, screen 3a, and JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = -10 to -30 V dc, $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. Power shall be applied to the device to achieve a junction temperature, $T_J$ = +175°C minimum and a minimum $P_D$ = 75 percent of $P_T$ maximum rated as defined in 1.3 herein.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). The thermal impedance limit used in screen 3c of 4.3 herein and table I shall comply with the thermal impedance graph on figures 10, 11, and 12 (less than or equal to the curve value at the same $t_H$ time) and shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.



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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA of MIL-PRF-19500 (JANS) and 4.4.2.1 herein. Delta measurements shall be in accordance with table III herein. See 4.4.2.2 herein JAN, JANTX, and JANTXV group B testing. Delta measurements shall be in accordance with table III herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = -40 V dc ±1 V, adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. |
| B5 | 1027 | (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample). $V_{CB}$ = -10 V dc; $P_D \geq$ 100 percent of maximum rated $P_T$ (see 1.3) $T_A \leq$ +35°C.<br><br>Option 1: 96 hours minimum, sample size in accordance with table E-VIA of MIL-PRF-19500, adjust $P_D$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hours, sample size = 45, c = 0; adjust $P_D$ to achieve $T_J$ = +225°C minimum. |
| B6 | 3131 | See 4.5.2. |

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIC of MIL-PRF-19500. Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new assembly lot option is exercised, the failed assembly lot shall be scrapped.

| Step | Method | Conditions |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = -10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Delta measurements shall be in accordance with table III herein, and only apply to subgroup C6.

4.4.3.1 Quality level JANS (see table E-VII of MIL-PRF-19500).

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; (method 2036 MIL-STD 750 not applicable for U3 devices). |
| C6 | 1026 | 1,000 hours at $V_{CB}$ = -10 V dc; power shall be applied to achieve $T_J$ = +150°C minimum and a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. n = 45, c = 0. |

4.4.3.2 Quality levels JAN, JANTX and JANTXV (see table E-VII of MIL-PRF-19500).

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; not applicable for U3 devices. |
| C5 | 3131 | $R_{\theta JA}$ for TO-205 (see 1.3), $R_{\theta JC}$ for U3 (see 1.3). |
| C6 | | Not applicable. |

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2 herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table IV herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta measurements shall be in accordance with table III herein.

4.5 Methods of inspection and test. Methods of inspection and test shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.



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4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be -500 mA minimum dc.

b. Collector to emitter voltage magnitude shall be -10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference temperature measuring point shall be within the range +25°C ≤ $T_R$ ≤ +35°C. The chosen reference temperature shall be recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. See 1.3 for maximum limit of $R_{\theta JC}$.

* 4.5.3 Displacement damage characterization. For RHA devices, each supplier shall perform a displacement damage characterization. The characterization shall demonstrate exposure versus data and does not indicate pass or fail criteria. The exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2E+12 n/cm$^2$ plus two additional fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest fluence level, a single data point may be sufficient.

Alternate package options may be substituted for characterization. The displacement damage characterization data shall be made available from the manufacturer.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical examination 3/ | 2071 | | | | | |
| Solderability 3/ 4/ | 2026 | | | | | |
| Resistance to solvents 3/ 4/ 5/ | 1022 | | | | | |
| Salt atmosphere (corrosion)(For laser marked devices only) 4/ | 1041 | n = 6 devices, c = 0 | | | | |
| Temp cycling 3/ 4/ | 1051 | Test condition C, 25 cycles | | | | |
| Hermetic seal 4/ 6/<br>Fine leak<br>Gross leak | 1071 | | | | | |
| Electrical measurements 4/ | | Group A, subgroup 2 | | | | |
| Bond strength 3/ 4/ | 2037 | Precondition:<br>$T_A$ = +250°C at t = 24 hrs. or<br>$T_A$ = +300°C at t = 2 hrs. | | | | |
| De-cap internal visual | 2075 | n = 4, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 7/ | 3131 | See 4.3.3 | $Z_{\theta JX}$ | | | °C/W |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D, $I_C$ = -100 mA dc; $I_B$ = 0, pulsed (see 4.5.1) | $V_{(BR)CEO}$ | -80 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C, $V_{CE}$ = -60 V dc; $V_{BE}$ = 0 | $I_{CES1}$ | | -1.0 | µA dc |
| Collector to emitter cutoff current | 3041 | Bias condition C, $V_{CE}$ = -100 V dc; $V_{BE}$ = 0 | $I_{CES2}$ | | -1.0 | mA dc |
| Collector to emitter cutoff current | 3041 | Bias condition D, $V_{CE}$ = -40 V dc; $I_B$ = 0 | $I_{CEO}$ | | -50 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -4 V dc; $I_C$ = 0 | $I_{EBO1}$ | | -1.0 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -5.5 V dc; $I_C$ = 0 | $I_{EBO2}$ | | -1.0 | mA dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -50 mA dc, pulsed (see 4.5.1) | $h_{FE1}$ | | | |
| 2N5151, L, and U3<br>2N5153, L, and U3 | | | | 20<br>50 | | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -2.5 A dc, pulsed (see 4.5.1) | $h_{FE2}$ | | | |
| 2N5151, L, and U3<br>2N5153, L, and U3 | | | | 30<br>70 | 90<br>200 | |
| Forward current transfer ratio | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -5 A dc, pulsed (see 4.5.1) | $h_{FE3}$ | | | |
| 2N5151, L, and U3<br>2N5153, L, and U3 | | | | 20<br>40 | | |
| Base-emitter voltage (non-saturated) | 3066 | Test condition B, $V_{CE}$ = -5 V dc; $I_C$ = -2.5 A dc, pulsed (see 4.5.1) | $V_{BE}$ | | -1.45 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A, $I_C$ = -2.5 A dc; $I_B$ = -250 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | -1.45 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A, $I_C$ = -5 A dc; $I_B$ = -500 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | -2.2 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -2.5 A dc; $I_B$ = -250 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | -0.75 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -5 A dc; $I_B$ = -500 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | -1.5 | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{CE}$ = -60 V dc; $V_{BE}$ = +2 V dc | $I_{CEX}$ | | -25 | µA dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 - continued | | | | | | |
| Low temperature operation | | $T_C$ = -55°C | | | | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -2.5 A dc; pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N5151, L, and U3 | | | | 15 | | |
| 2N5153, L, and U3 | | | | 25 | | |
| Subgroup 4 | | | | | | |
| Common-emitter, small-signal, short-circuit, forward-current transfer ratio | 3206 | $V_{CE}$ = -5 V dc; $I_C$ = -100 mA dc; f = 1 KHz | $h_{fe}$ | | | |
| 2N5151, L, and U3 | | | | 20 | | |
| 2N5153, L, and U3 | | | | 50 | | |
| Magnitude of common-emitter, small-signal short-circuit, forward-current, transfer ratio | 3306 | $V_{CE}$ = -5 V dc; $I_C$ = -500 mA dc, f = 10 MHz | $\lvert h_{fe} \rvert$ | | | |
| 2N5151, L, and U3 | | | | 6 | | |
| 2N5153, L, and U3 | | | | 7 | | |
| Open-circuit output capacitance | 3236 | $V_{CB}$ = -10 V dc; $I_E$ = 0, f = 1 MHz | $C_{obo}$ | | 250 | pF |
| Switching time | | $I_C$ = -5 A dc; $I_{B1}$ = -500 mA dc | $t_{on}$ | | 0.5 | µs |
| | | $I_{B2}$ = -500 mA dc | $t_s$ | | 1.4 | µs |
| | | $V_{BE(off)}$ = -3.7 V dc | $t_f$ | | 0.5 | µs |
| | | $R_L$ = 6 Ω, (see figure 14) | $t_{off}$ | | 1.5 | µs |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area (dc) | 3051 | Pre-pulse condition for each test: $T_C$ = +25°C, (see figure 15)<br>Pulse condition for each test: $t_p$ = 1 sec. 1 cycle $T_C$ = +25°C | | | | |
| Test # 1 | | $V_{CE}$ = -5.0 V dc, $I_C$ = -2 A dc for TO-205<br>$V_{CE}$ = -5.8 V dc, $I_C$ = -2 A dc for U3 | | | | |
| Test # 2 | | $V_{CE}$ = -32 V dc, $I_C$ = -310 mA dc for TO-205<br>$V_{CE}$ = -32 V dc, $I_C$ = -360 mA dc for U3 | | | | |
| Test # 3 | | $V_{CE}$ = -80 V dc, $I_C$ = -12.5 mA dc for TO-205<br>$V_{CE}$ = -80 V dc, $I_C$ = -14.5 mA dc for U3 | | | | |
| Safe operating area (unclamped inductive) | 3053 | Condition C $T_C$ = +25°C;<br>$R_{BB1}$ = 10 Ω;<br>$R_{BB2}$ = 100 Ω; L = 0.3 mH;<br>RL = 0.1 Ω; $V_{CC}$ = -10 V dc;<br>$V_{BB1}$ = -10 V dc; $V_{BB2}$ = 4 V dc;<br>$I_C$ = -10 A dc (see figure 13) | | | | |
| End point electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ Hermetic seal test is an end-point to temperature cycling in addition to electrical measurements.
* 7/ For end-point measurements, this test is required for the following subgroups:
Group B, steps 2 and 3 (JAN, JANTX, and JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



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* TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 4/ | | | | | | |
| Neutron irradiation | 1017 | Neutron exposure $V_{CES}$ = 0V | | | | |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D, $I_C$ = -100 mA dc; $I_B$ = 0, pulsed (see 4.5.1) | $V_{(BR)CEO}$ | -80 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C, $V_{CE}$ = -60 V dc; $V_{BE}$ = 0 | $I_{CES1}$ | | -2.0 | µA dc |
| Collector to emitter cutoff current | 3041 | Bias condition C, $V_{CE}$ = -100 V dc; $V_{BE}$ = 0 | $I_{CES2}$ | | -2.0 | mA dc |
| Collector to emitter cutoff current | 3041 | Bias condition D, $V_{CE}$ = -40 V dc; $I_B$ = 0 | $I_{CEO}$ | | -100 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -4 V dc; $I_C$ = 0 | $I_{EBO1}$ | | -2.0 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -5.5 V dc; $I_C$ = 0 | $I_{EBO2}$ | | -2.0 | mA dc |
| Forward-current transfer ratio<br>2N5151<br>2N5153 | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -50 mA dc | [$h_{FE1}$] 5/ | <br>[10]<br>[25] | | |
| Forward-current transfer ratio<br>2N5151<br>2N5153 | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -2.5 A dc, pulsed | [$h_{FE2}$] 5/ | <br>[15]<br>[35] | <br>90<br>200 | |
| Forward-current transfer ratio<br>2N5151<br>2N5153 | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -5 A dc, pulsed | [$h_{FE3}$] 5/ | <br>[10]<br>[20] | | |
| Base-emitter voltage (non-saturated) | 3066 | Test condition B, $V_{CE}$ = -5 V dc; $I_C$ = -2.5 A dc, pulsed (see 4.5.1) | $V_{BE}$ | | -1.67 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A, $I_C$ = -2.5 A dc; $I_B$ = -250 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | -1.67 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A, $I_C$ = -5 A dc; $I_B$ = -500 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | -2.53 | V dc |

See footnotes at end of table.



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* TABLE II. Group D inspection. Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 - Continued 4/ | | | | | | |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -2.5 A dc; $I_B$ = -250 mA dc; pulsed | $V_{CE(sat)1}$ | | -0.86 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -5 A dc; $I_B$ = -500 mA dc; pulsed | $V_{CE(sat)2}$ | | -1.73 | V dc |
| Subgroup 2. | | | | | | |
| * Total dose irradiation | 1019 | Condition A, Gamma exposure , $V_{CES}$ = -64 V | | | | |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D, $I_C$ = -100 mA dc; $I_B$ = 0, pulsed (see 4.5.1) | $V_{(BR)CEO}$ | -80 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C, $V_{CE}$ = -60 V dc; $V_{BE}$ = 0 | $I_{CES1}$ | | -2.0 | µA dc |
| Collector to emitter cutoff current | 3041 | Bias condition C, $V_{CE}$ = -100 V dc; $V_{BE}$ = 0 | $I_{CES2}$ | | -2.0 | mA dc |
| Collector to emitter cutoff current | 3041 | Bias condition D, $V_{CE}$ = -40 V dc; $I_B$ = 0 | $I_{CEO}$ | | -100 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -4 V dc; $I_C$ = 0 | $I_{EBO1}$ | | -2.0 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = -5.5 V dc; $I_C$ = 0 | $I_{EBO2}$ | | -2.0 | mA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -50 mA dc | [$h_{FE1}$] 5/ | | | |
| 2N5151 | | | | [10] | | |
| 2N5153 | | | | [25] | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -2.5 A dc | [$h_{FE2}$] 5/ | | | |
| 2N5151 | | | | [15] | 90 | |
| 2N5153 | | | | [35] | 200 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -5 V dc; $I_C$ = -5 A dc. pulsed | [$h_{FE3}$] 5/ | | | |
| 2N5151 | | | | [10] | | |
| 2N5153 | | | | [20] | | |



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TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued. | | | | | | |
| Base-emitter voltage (non-saturated) | 3066 | Test condition B, $V_{CE}$ = -5 V dc; $I_C$ = -2.5 A dc, pulsed (see 4.5.1) | $V_{BE}$ | | -1.67 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A, $I_C$ = -2.5 A dc; $I_B$ = -250 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | -1.67 | V dc |
| Base-emitter saturation voltage | 3066 | Test condition A, $I_C$ = -5 A dc; $I_B$ = -500 mA dc; pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | -2.53 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -2.5 A dc; $I_B$ = -250 mA dc | $V_{CE(sat)1}$ | | -0.86 | V dc |
| Collector-emitter saturation voltage | 3071 | $I_C$ = -5 A dc; $I_B$ = -500 mA dc | $V_{CE(sat)2}$ | | -1.73 | V dc |

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to all device types unless otherwise noted.
4/ Subgroup 1 is an optional test and shall be specified on the contract when required.
5/ See method 1019 of MIL-STD-750 for how to determine [$h_{FE}$] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation $h_{FE}$. Notice that [$h_{FE}$] is not the same as $h_{FE}$ and cannot be measured directly. The [$h_{FE}$] value can never exceed the pre-radiation minimum $h_{FE}$ that it is based upon.



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TABLE III. Groups B, C, and E delta and electrical measurements. 1/ 2/ 3/ 4/

| Steps | Inspection | MIL-STD-750 | | Symbol | Limits | Limits | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Forward - current transfer ratio | 3076 | $I_C$ = -2.5 A dc; $V_{CE}$ = -5 V dc, pulsed (see 4.5.1). | $\Delta h_{FE2}$ | ±20 percent change from initial reading. | | |

1/ The delta measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows: Subgroups 4 and 5, see table III herein, step 1.
2/ The delta measurements for 4.4.2.2 (JAN, JANTX and JANTXV) for all steps; see table III herein, step 1.
3/ The delta measurements for table E-VII of MIL-PRF-19500 are as follows: Subgroup 6, see table III herein, step 1.
4/ The delta measurements for table E-IX of MIL-PRF-19500 and table III herein are as follows: Subgroups 1 and 2, see table III herein, step 1.



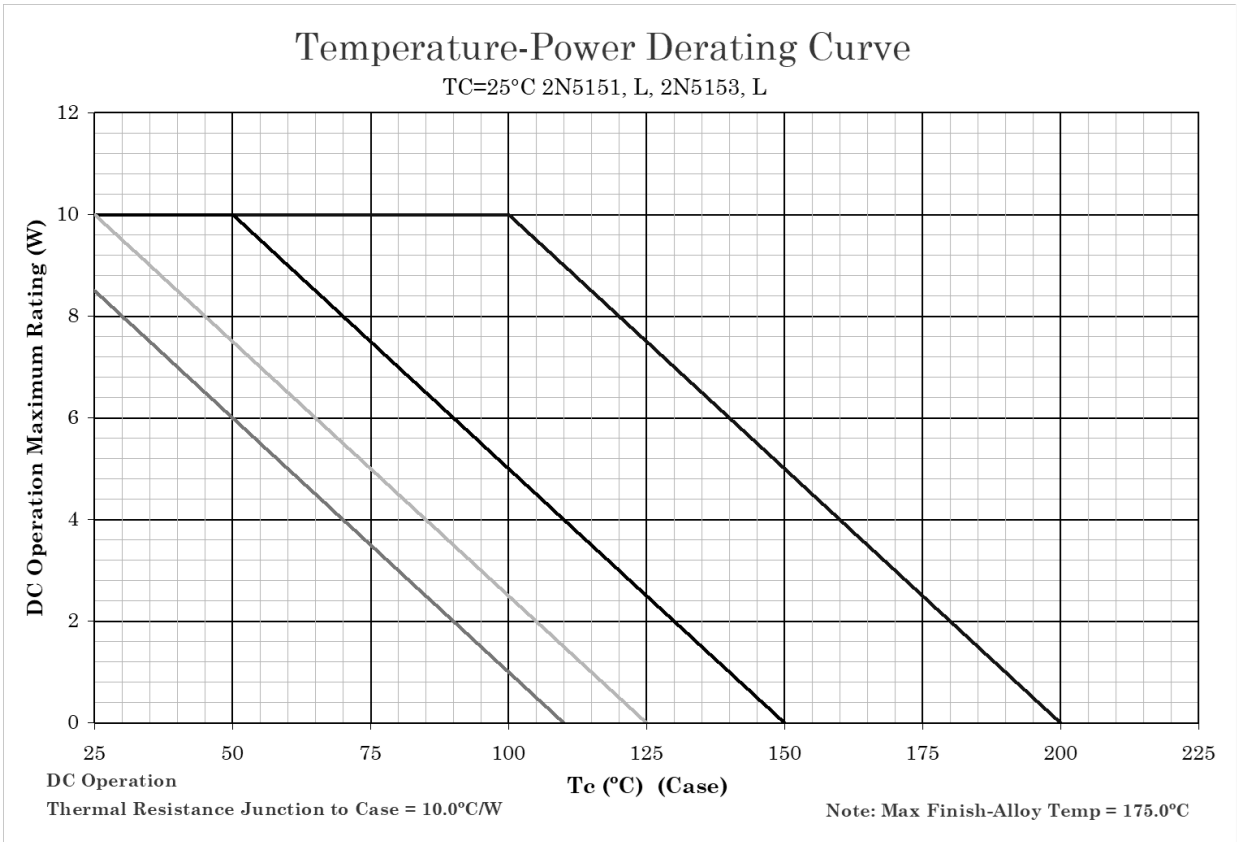
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TABLE IV. Group E inspection (all quality levels) – for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 and table III herein | |
| Subgroup 2 | | | 45 devices c = 0 |
| Intermittent life | 1037 | $V_{CB}$ = -10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C | |
| Electrical measurements | | See table I, subgroup 2 and table III herein | |
| Subgroup 4 | | | |
| Thermal impedance curves | | See table E-IX of MIL-PRF-19500, group E, subgroup 4. | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 8 | | | 45 devices c = 0 |
| Reverse stability | 1033 | Condition B | |
| * Subgroup 12 | | | |
| Neutron Irradiation | 1017 | See 4.5.4 | |



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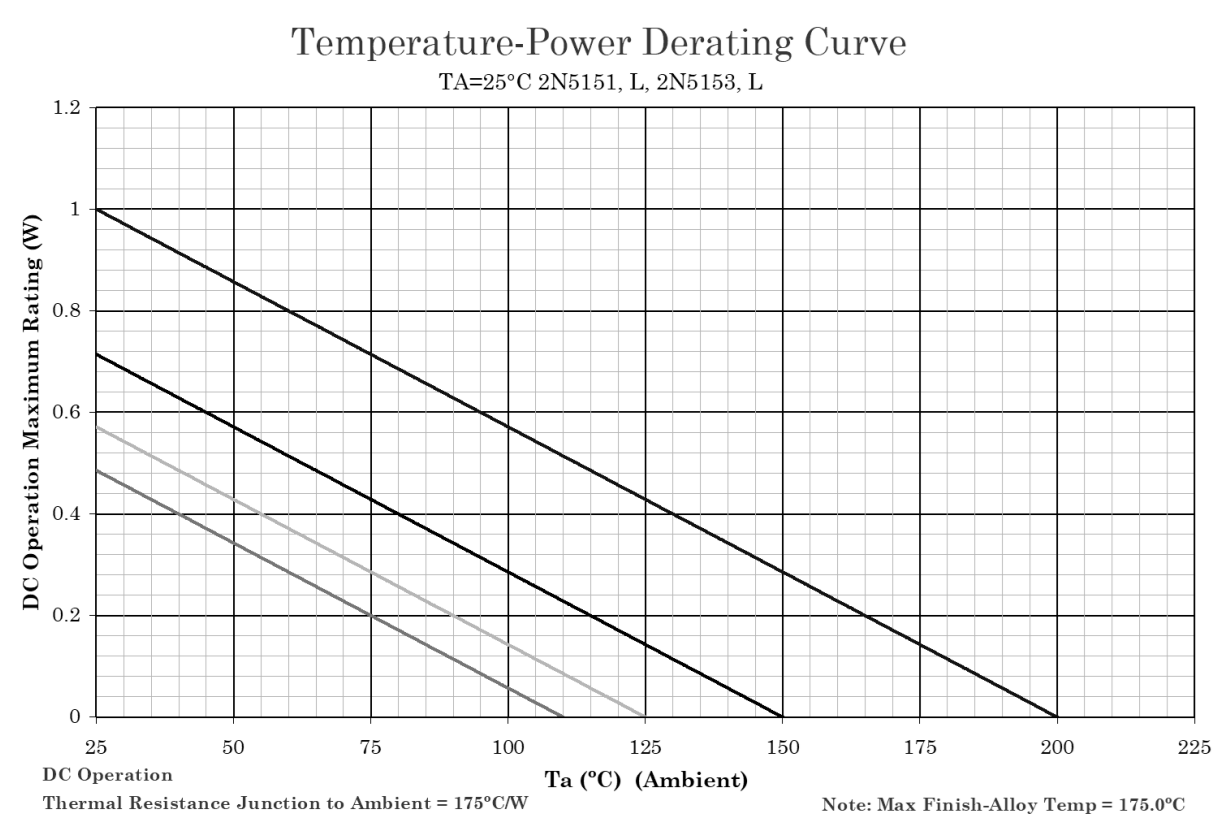

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 6. Temperature-power derating graph, TO-205, case temperature.



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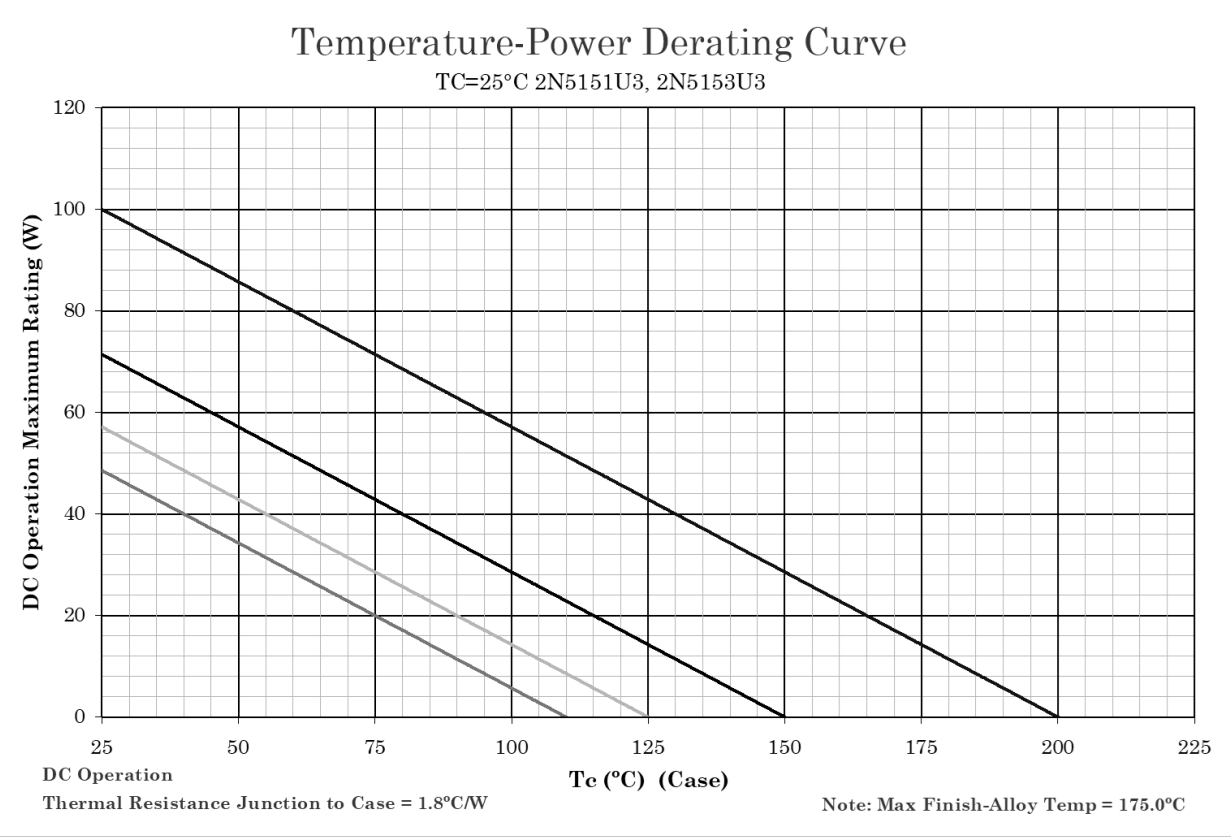

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 7. Temperature-power derating graphs, TO-205 ambient temperature.



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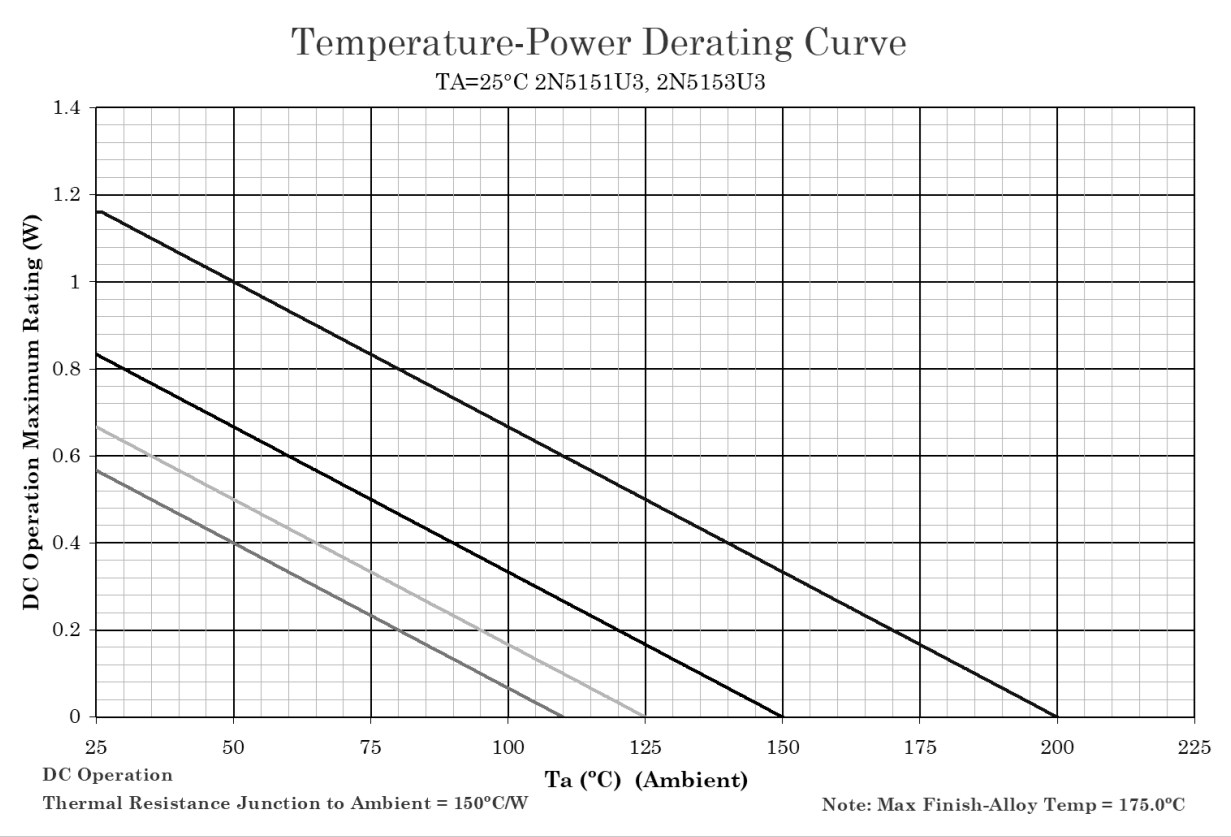

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 8. Temperature-power derating graph, U3 package, case temperature.



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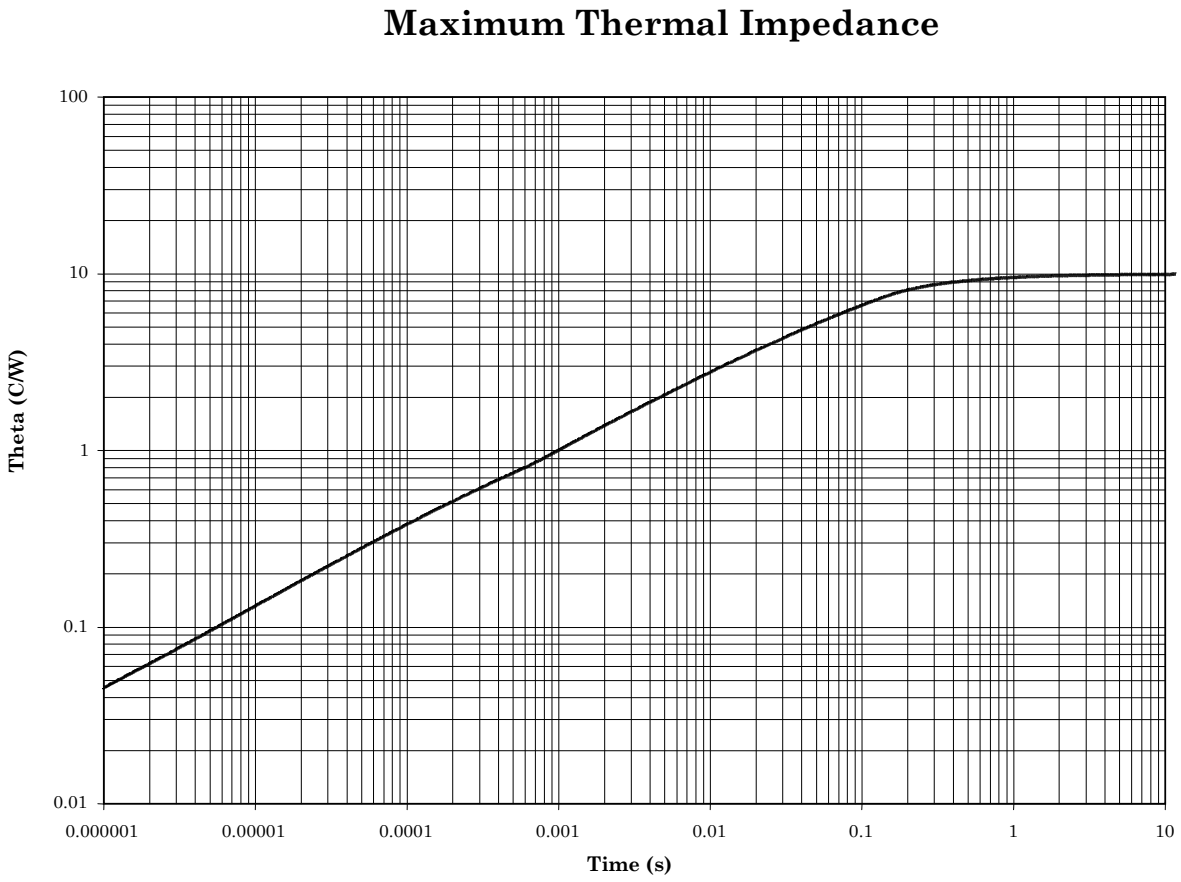

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ +125°C, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 9. Temperature-power derating graph, U3 package, ambient temperature.



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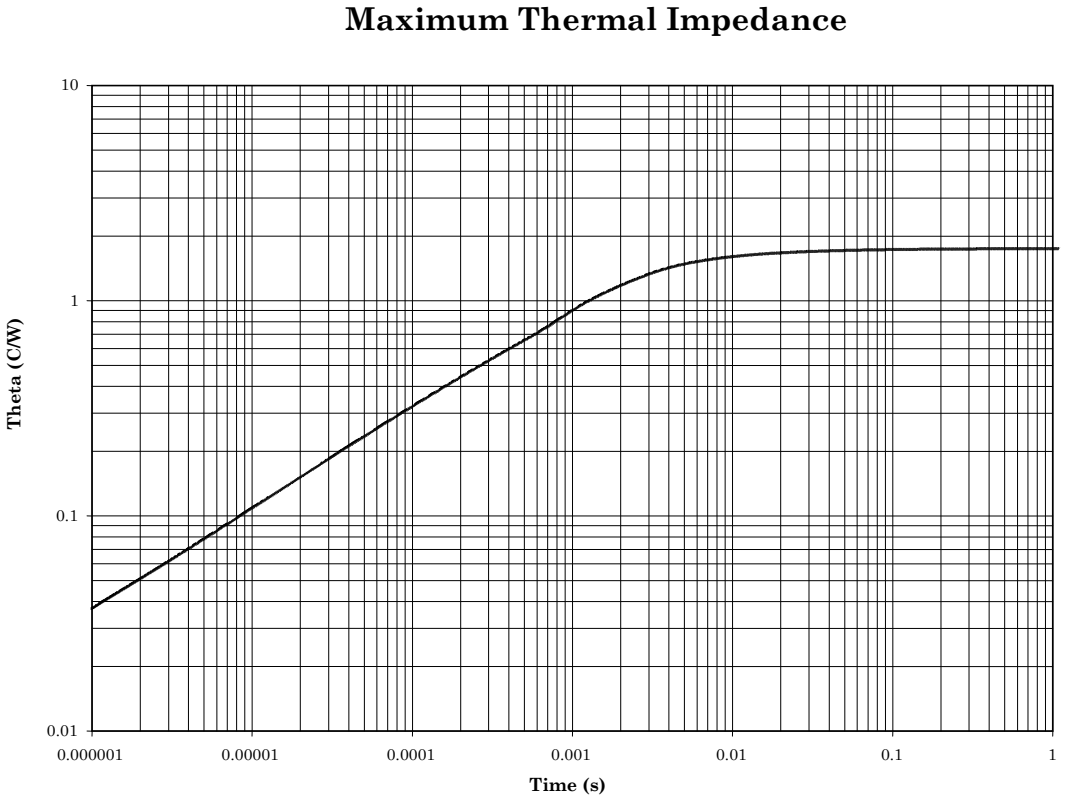



2N5151, 2N5151L, 2N5153, and 2N5153L at $T_C$ = +25°C, $R_{\theta JC}$ = 10°C/W.

FIGURE 10. Thermal impedance graph, TO-205 package at case temperature.



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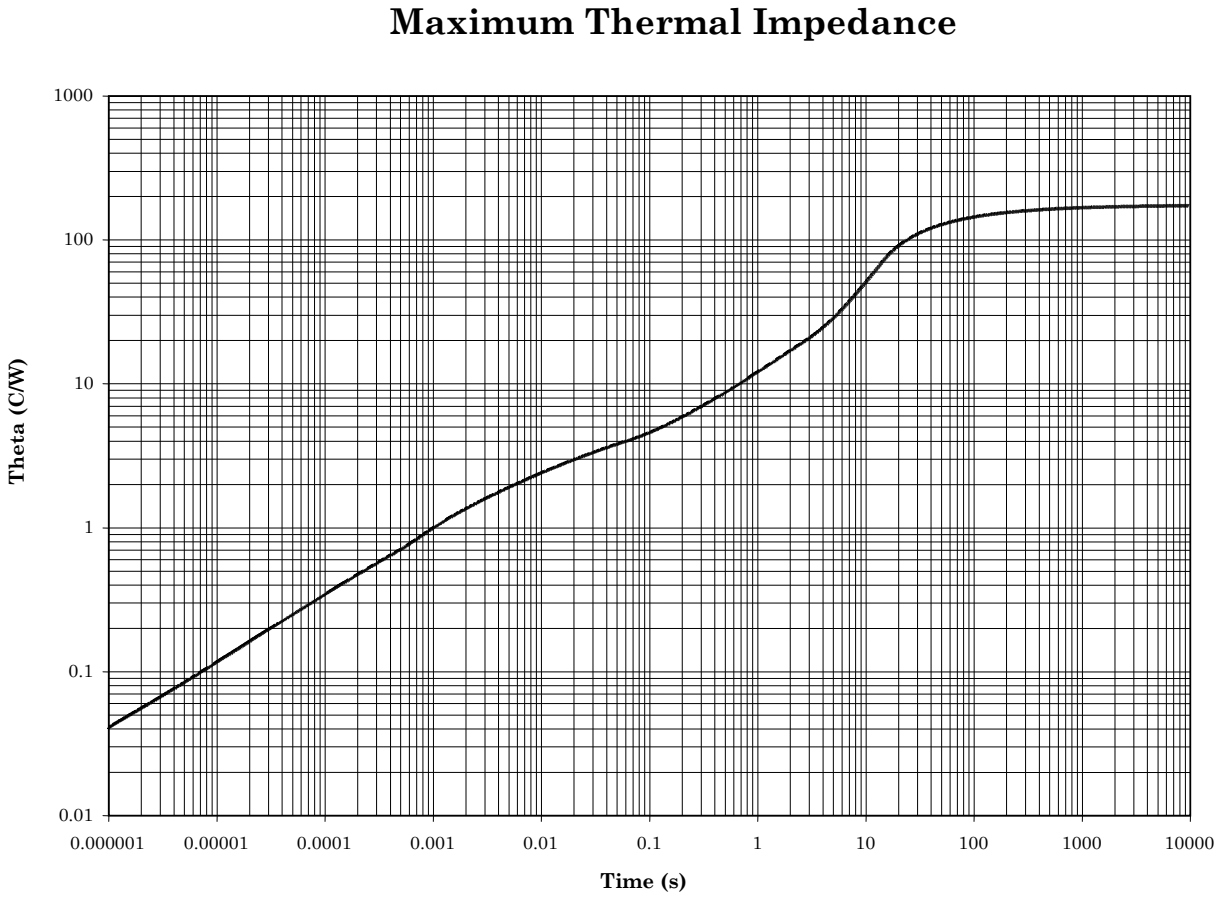



2N5151U3 and 2N5153U3 at $T_C$ = +25°C, $R_{\theta JC}$ = 1.75°C/W.

FIGURE 11. Thermal impedance graph, U3 package at case temperature.



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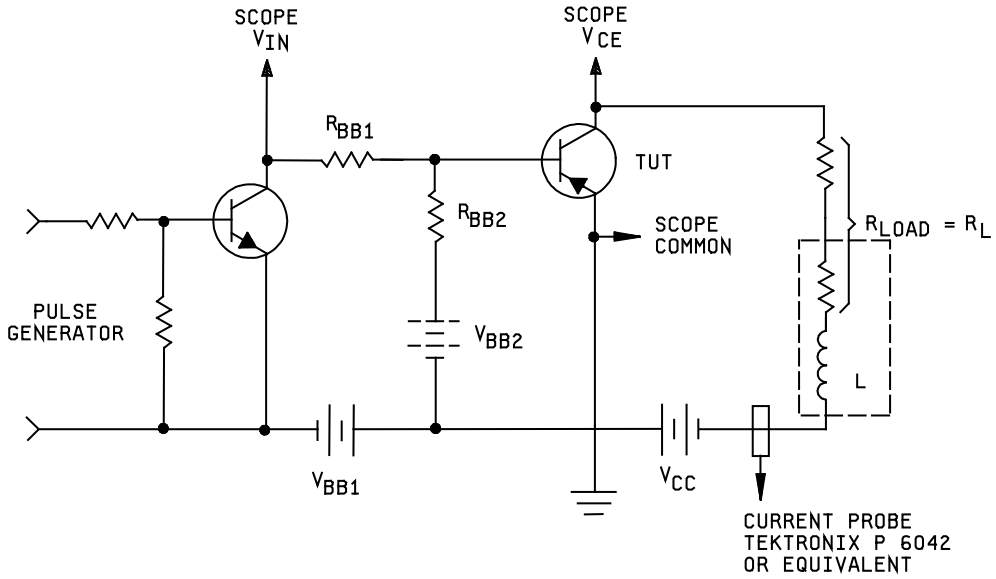



2N5151, 2N5151L, 2N5153, and 2N5153L at $T_A$ = +25°C, $R_{\theta JA}$ = 175°C/W.

FIGURE 12. Thermal impedance graph, TO-205 package at ambient temperature.



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Check the source to verify that this is the current version before use.

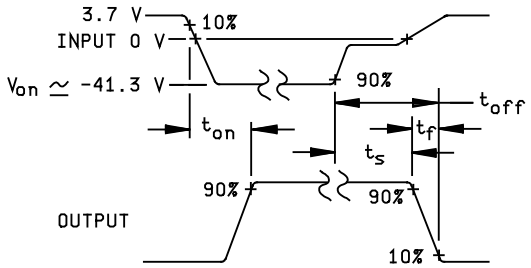


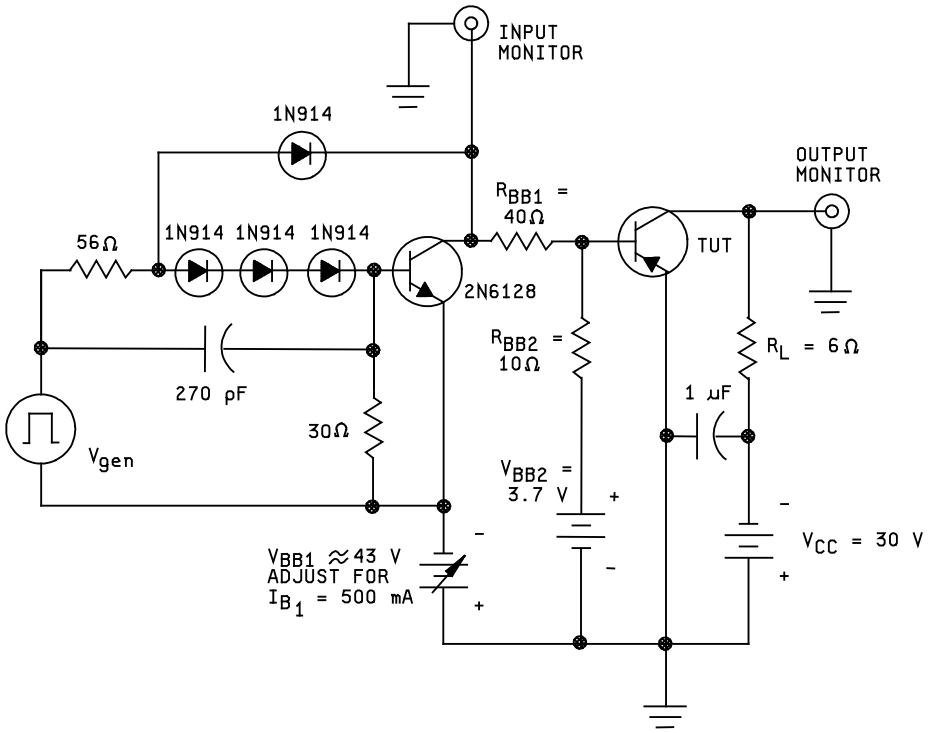

$R_{BB1}$ = 10 Ω
$R_{BB2}$ = 100 Ω
L = 0.3 mH
$R_L$ = 0.1 Ω
$V_{CC}$ = -10 V dc
$I_C$ = -10 A
$V_{BB1}$ = -10 V dc
$V_{BB2}$ = 4 V dc

FIGURE 13. Unclamped inductive load energy test circuit.



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NOTES:

1. $V_{gen}$ is -30 pulse (from 0 V) into a 50 ohm termination.
2. The $V_{gen}$ waveform is supplied by a generator with the following characteristics: $t_r \leq 15$ ns, $t_f = 15$ ns, $Z_{OUT}$ = 50 ohm, duty cycle ≤ 2 percent.
3. Waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $R_{IN} \geq 10$ MΩ, $C_{IN} \leq 11.5$ pF.
4. Resistors shall be noninductive types.
5. The dc power supplies may require additional bypassing in order to minimize ringing.
6. An equivalent circuit may be used.

FIGURE 14. Switching time test circuit.



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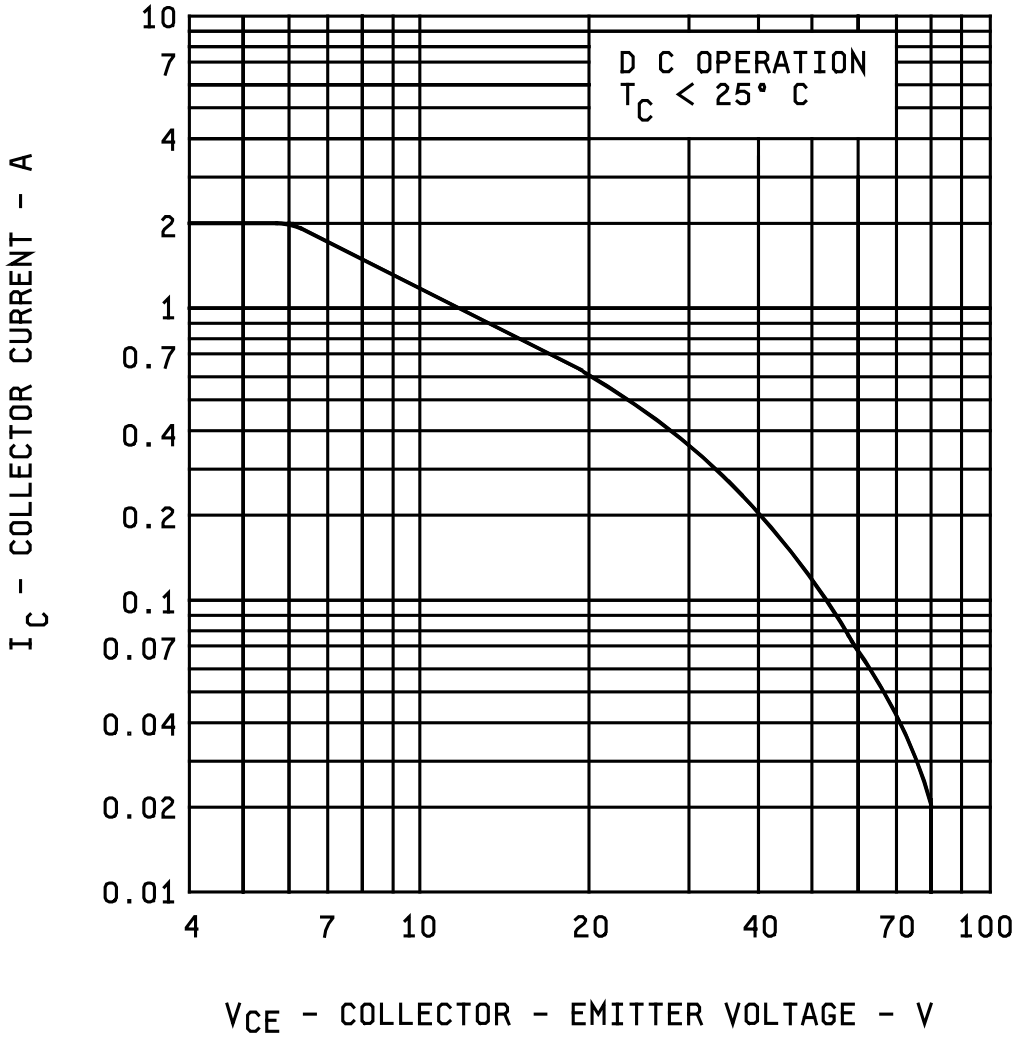



FIGURE 15. Maximum safe operating area.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.2).

d. The complete PIN, see 1.5 and 6.5.

e. For acquisition of RHA designed devices, table II herein, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it will be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.



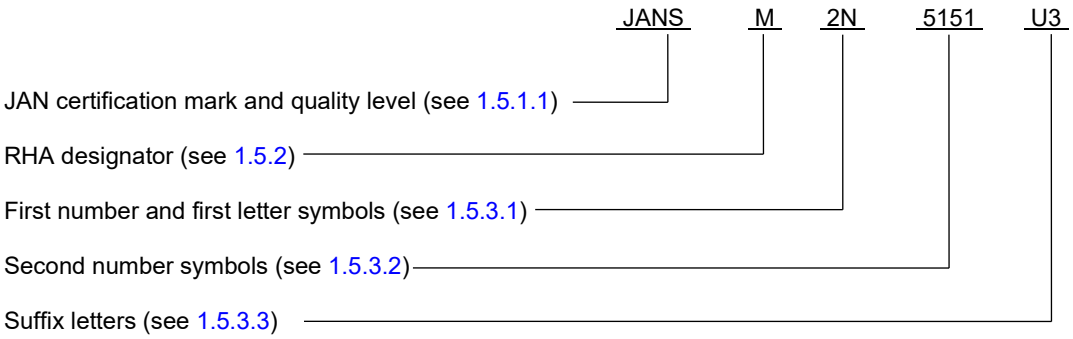
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* 6.4. Suppliers and PINs of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCB2N5151) will be identified on the QML.

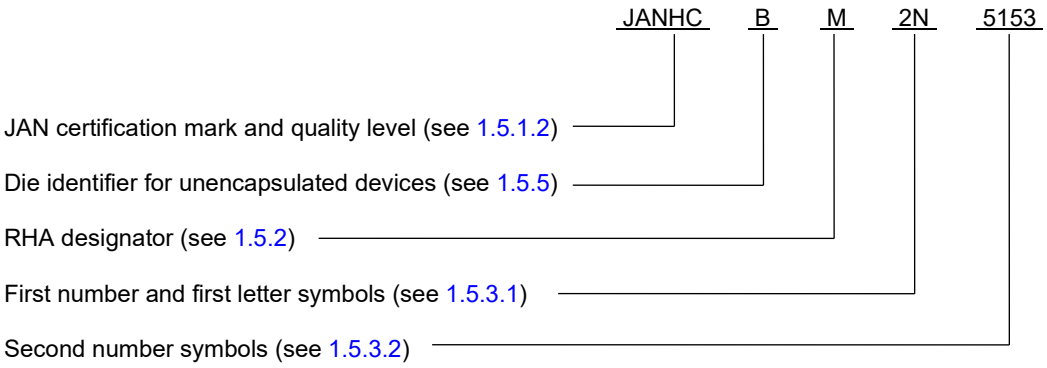
| JANHC and JANKC ordering information | | | |
|---|---|---|---|
| PIN | Manufacturer | | |
| | 34156 | 43611 | 52GC4 |
| 2N5151<br>2N5153 | JANHCB2N5151<br>JANHCB2N5153 | JANHCC2N5151<br>JANHCC2N5153 | JANHCD2N5151<br>JANHCD2N5153 |
| 2N5151<br>2N5153 | JANKCB2N5151<br>JANKCB2N5153 | JANKCC2N5151<br>JANKCC2N5153 | JANKCD2N5151<br>JANKCD2N5153 |

6.5 PIN construction example.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.




6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.






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6.6 List of PINs.

6.6.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for type 2N5151 and 2N5153. | | | |
|---|---|---|---|
| JAN2N5151 | JANTX2N5151 | JANTXV#2N5151 | JANS#2N5151 |
| JAN2N5151L | JANTX2N5151L | JANTXV#2N5151L | JANS#2N5151L |
| JAN2N5153 | JANTX2N5153 | JANTXV#2N5153 | JANS#2N5153 |
| JAN2N5153L | JANTX2N5153L | JANTXV#2N5153L | JANS#2N5153L |
| JAN2N5151U3 | JANTX2N5151U3 | JANTXV#2N5151U3 | JANS#2N5151U3 |
| JAN2N5153U3 | JANTX2N5153U3 | JANTXV#2N5153U3 | JANS#2N5153U3 |

(1) The number sign (#) represents one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

* 6.6.2 PINs for unencapsulated devices (die). The following is a list of possible PINs for unencapsulated devices available on this specification sheet.

| Quality level HC | Quality level KC |
|---|---|
| JANHCB#2N5151 | JANKCB#2N5151 |
| JANHCB#2N5153 | JANKCB#2N5153 |
| JANHCC#2N5151 | JANKCC#2N5151 |
| JANHCC#2N5153 | JANKCC#2N5153 |
| JANHCD#2N5151 | JANKCD#2N5151 |
| JANHCD#2N5153 | JANKCD#2N5153 |

(1) The number sign (#) represents one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

* 6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - MI

Preparing activity:
DLA - CC

(Project 5961-2022-033)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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